



Exchange rate expectations and explainable AI[☆]

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ARTICLE INFO

JEL classification:

F31
F33
E52

Keywords:

Exchange rate expectations
Monetary policy
Survey data
UIP
XGBoost

ABSTRACT

This paper examines the determinants of long-term exchange rate expectations using survey data for advanced and emerging economies. We complement standard regressions with explainable machine learning (XGBoost with SHAP values) to capture nonlinearities and interactions. We find substantial differences across country groups, with inflation forecast errors and uncertainty more important in emerging markets. Monetary policy shocks become relevant once nonlinearities are accounted for. Overall, the results highlight state-dependent expectation formation and the value of explainable machine learning as a complement to standard methods.

1. Introduction

Due to its relevance for foreign direct investments and the effectiveness of monetary policy, the dynamics of long-term exchange rate expectations represent a crucial issue in international macroeconomics. The empirical exchange rate literature, dating back to the contribution of Meese and Rogoff (1983), has documented the remarkable difficulty in forecasting exchange rates with macroeconomic fundamentals (Rossi, 2013). Although recent advances have shown that monetary policy surprises, rather than observed interest rate differentials, help to reconcile long-horizon expectations with uncovered interest parity (Czudaj and Rezaei Khavas, 2026), the empirical framework remains largely linear.

However, expectation formation in currency markets is unlikely to be purely linear. Professional forecasters operate in environments characterized by regime changes, nonlinear monetary transmission, asymmetric responses to inflation surprises, and state-dependent reactions to uncertainty. The (delayed) overshooting literature (Dornbusch, 1976; Eichenbaum and Evans, 1995; Müller et al., 2024) suggests dynamic asymmetries and nonlinear adjustments. Similarly, safe-haven dynamics (Ranaldo and Söderlind, 2010; Habib and Stracca, 2012) indicate that the impact of uncertainty shocks may depend on global risk regimes. These considerations raise the question of whether linear panel regressions are sufficiently flexible to capture the structure underlying long-term exchange rate expectations.

A preceding study by Czudaj and Rezaei Khavas (2026) demonstrates that isolating exogenous monetary policy shocks via forecast errors restores UIP-consistent behavior in survey expectations. It further shows that inflation differentials dominate at longer horizons, whereas uncertainty shocks play only a limited role. However, these findings leave open an important possibility: the effects of monetary policy and uncertainty may be nonlinear, interaction-driven, or threshold-dependent. For instance, monetary tightening may affect expectations differently in high-inflation versus low-inflation environments, or in Emerging Market and Developing

[☆] Thanks for valuable comments on a previous draft of the paper are due to two anonymous reviewers.

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Economies (EMDE) compared to Advanced Economies (AE). Similarly, uncertainty shocks may matter only during crisis periods or when combined with monetary surprises.

To address these limitations, this study builds on the approach proposed by Berger (2023, 2025) and applies explainable artificial intelligence techniques to allow for nonlinearities, interaction effects, and complex hierarchical structures. Specifically, we employ Extreme Gradient Boosting (XGBoost; Chen and Guestrin, 2016) in combination with SHAP values (Lundberg and Lee, 2017), thereby enabling a direct comparison between widely used fixed-effects panel regressions and state-of-the-art machine learning methods. While linear fixed effects models assume additive and homogeneous effects, XGBoost can adapt to regime-specific dynamics, such as asymmetric responses to shocks or threshold effects, which are likely to differ systematically between advanced and emerging markets. This flexibility allows the model to uncover patterns that may be missed by conventional econometric approaches, thereby enhancing both predictive accuracy and interpretability through methods like SHAP analysis.

Applying XGBoost to a panel of 29 AE and EMDE currencies allows us to revisit core questions in the exchange rate expectations literature within a flexible, data-driven framework. We examine whether monetary policy surprises remain the dominant determinant of long-term exchange rate expectations once nonlinearities, interactions, and heterogeneity across countries are explicitly accounted for. At the same time, we investigate whether the role of uncertainty shocks may be masked by state-dependent effects that materialize only under specific conditions, such as in economies with weaker fundamentals or during periods of elevated global risk. By addressing these questions, we aim to reconcile discrepancies between theory and empirical evidence and to provide a more nuanced view of expectation formation. Beyond predictive gains, the use of explainable machine learning tools enables economic interpretation of nonlinear effects, helping bridge the gap between data-driven modeling and macroeconomic theory to provide a deeper understanding of expectation formation.

The remainder is organized as follows: Section 2 introduces the data, Section 3 describes the empirical strategy, Section 4 discusses the findings, and Section 5 concludes.

2. Data

The empirical analysis is based on a balanced annual panel covering 29 currencies of AE and EMDE over the period 2008–2021.¹ Bilateral exchange rates are defined in units of domestic currency per US dollar, such that an increase reflects a depreciation of the domestic currency. The sample comprises a broad cross-section of countries, allowing for a comparison between AE and more vulnerable EMDE.

Long-term exchange rate expectations are proxied using survey data from FX4casts (<https://fx4casts.com/>). The survey collects monthly forecasts from professional analysts employed at major financial institutions and research institutes.² We focus on end-of-period forecasts at fixed horizons of three and four years ahead. To align survey expectations with macroeconomic data at annual frequency, we construct yearly forecast revisions as percentage changes based on the January vintage of each year. This measure captures how professional forecasters update their long-term exchange rate views over the course of a year.

To explain forecast revisions, we include traditional macroeconomic fundamentals expressed as differentials relative to the United States (US). These comprise realized and forecasted short-term nominal interest rates (three-month rates), real GDP growth rates, and inflation rates. For the latter three, we consider the differentials of forecasts relative to the US and of the corresponding forecast errors. The forecasts are also drawn from FX4casts, while realized macroeconomic data for GDP growth and inflation are taken from the IMF World Economic Outlook database (<https://www.imf.org/en/publications/sprolls/world-economic-outlook-databases>). All variables are expressed in annual frequency to ensure consistency with the constructed expectation updates. In this context, an important issue is the separation of anticipated and unanticipated interest rate movements. Rather than relying solely on observed interest rate differentials, which may already be priced into expectations, we identify monetary policy surprises using forecast errors (Furceri et al., 2018; Czudaj and Rezaei Khavas, 2026).

Unexpected interest rate changes are computed as the difference between realized interest rates and the corresponding one-year-ahead forecasts made in the previous January survey. This procedure isolates the component of interest rate movements that was not anticipated by professional forecasters. To mitigate concerns that interest rate forecast errors might reflect endogenous policy responses to macroeconomic conditions, we purge forecast errors of contemporaneous GDP growth and inflation forecast errors using country-specific regressions. The resulting residual component is interpreted as an exogenous monetary policy shock. Finally, shocks are expressed relative to the US to ensure consistency with the bilateral exchange rate perspective.

Uncertainty shocks are constructed using the World Uncertainty Index (WUI), a text-based indicator derived from country reports (<https://worlduncertaintyindex.com/>). Annual changes are computed as differences between fourth-quarter values of consecutive years. To also address potential endogeneity, we apply the same procedure as for monetary policy shocks and express uncertainty innovations relative to the US. As additional measures, we also consider the central bank independence (CBI) index by Romelli (2022) and the Debt-to-GDP from the IMF World Economic Outlook database.³

¹ The classification of economies into advanced and emerging refers to the IMF as follows: Australia, Canada, the Czech Republic, Denmark, the Euro Area, Japan, Korea, New Zealand, Norway, Singapore, Sweden, Switzerland, Taiwan, and the UK are considered as Advanced Economies (AE), and Argentina, Brazil, Chile, China, Colombia, Hungary, India, Indonesia, Mexico, Philippines, Poland, Russia, South Africa, Thailand, and Turkey as Emerging Market and Developing Economies (EMDE).

² While the empirical UIP literature often relies on forward rates or interest rate differentials as proxies for exchange rate expectations, such measures conflate expectations with time-varying risk premia (Frankel and Chinn, 1993; Engel, 1996) and therefore do not provide a clean measure of agents' beliefs. In contrast, survey-based expectations directly capture the subjective forecasts of professional market participants and allow us to isolate expectation formation from equilibrium pricing conditions.

³ Descriptive statistics are presented in Table A.1 in Appendix.

3. Empirical methodology

The fixed effect regression serves as a linear benchmark approach to identify the relevant explanatory variables that explain forecast revisions. The regression is given as follows:

$$y_{it} = \mathbf{X}_{it}\beta + \gamma_i + \delta_t + \varepsilon_{it}, \quad (1)$$

where y_{it} describes the target variable (exchange rate forecast revisions), γ_i and δ_t capture country- and year-specific unobserved effects, respectively, and $\mathbf{X}_{it}$ includes the key explanatory variables. In addition to linear panel data fixed-effects regression, we analyze individual-level determinants using explainable artificial intelligence as proposed in [Berger \(2025\)](#), employing Extreme Gradient Boosting (XGBoost) in combination with SHAP values.

3.1. Gradient boosting with XGBoost

As introduced in [Chen and Guestrin \(2016\)](#), XGBoost is an efficient implementation of gradient boosted decision trees, where an ensemble of weak learners is trained sequentially to minimize a regularized objective function. The use of XGBoost offers a significant advantage over traditional linear models by capturing complex, non-linear relationships and higher-order interactions among explanatory variables that are particularly relevant in heterogeneous samples of advanced and emerging market economies.

Formally, given a dataset $D = \{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$, the model prediction is defined as $\hat{y}_i = \sum_{k=1}^K f_k(x_i)$, $f_k \in \mathcal{F}$, where $\mathcal{F}$ denotes the space of regression trees. The training objective is $\mathcal{L} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$, with $\ell(\cdot)$ being a differentiable loss function and $\Omega(\cdot)$ a regularization term controlling model complexity. By optimizing this objective using second-order gradient information, XGBoost achieves high predictive performance while mitigating overfitting. The approach is used to predict forecast revisions (y_i) by a set of predictors (x_i) as introduced in [Section 2](#).

3.2. Model explainability via SHAP values

To interpret the predictions of the XGBoost model, we use *SHapley Additive exPlanations* (SHAP), a game-theoretic approach to feature attribution ([Lundberg and Lee, 2017](#)). SHAP values assign each feature a contribution to the model prediction by computing its average marginal contribution across all possible feature subsets.

Let x be the input as described in [Section 2](#), the model prediction can then be decomposed as

$$f(x) = \phi_0 + \sum_{j=1}^d \phi_j, \quad (2)$$

where ϕ_0 is the expected model output and ϕ_j denotes the SHAP value of feature j . These values satisfy properties of local accuracy, consistency, and additivity, ensuring a faithful explanation of the model behavior. As demonstrated in [Berger \(2025\)](#), in combination with tree-based models such as XGBoost, SHAP values can be efficiently computed and provide both local (instance-level) and global (dataset-level) interpretability, making them a suitable alternative to linear regression approaches. However, global SHAP values represent the average marginal contribution of a feature to predictions rather than statistical significance. A positive SHAP value indicates that a higher value of the feature increases the predicted outcome (on average), while a negative SHAP value indicates the opposite.

4. Empirical findings

[Table 1](#) reports and compares the empirical findings across fixed-effects regressions and the XGBoost model based on Shapley value decompositions. Although SHAP values capture the magnitude of a variable's contribution to prediction accuracy, they do not directly provide information about the direction of the association and are not inferential statistics, making them complementary to regression coefficients.⁴

Inflation forecast errors (CPI_Forecast_Error), which can be interpreted as shocks hitting the inflation rate, appear as a consistently strong determinant of changes in exchange rate expectations over both forecast horizons. Both the fixed-effects regression (1.488 for $h = 3$ and 1.865 for $h = 4$) and the XGBoost model, based on SHAP value decompositions (3.668 for $h = 3$ and 4.158 for $h = 4$), indicate a robust and strong association. Professional forecasters appear to anticipate that the domestic currency will depreciate against the US dollar in accordance with purchasing power parity ([Beckmann and Czudaj, 2025](#)) whenever inflation shocks for the domestic economy exceed those of the US. Specifically, we find that larger inflation shocks seem to lead to significantly stronger revisions in exchange rate expectations, particularly for emerging economies (fixed-effect: 1.472 and SHAP: 8.233). According to SHAP values, inflation forecast errors describe the dominant determinant in EMDE, while the importance seems to be weaker for AE. This is a reasonable finding as inflation rates are generally higher in EMDE than in AE and therefore also play a more relevant

⁴ It is to note, that we apply XGBoost exclusively to identify relevant determinants in an in-sample assessment. The resulting R^2 therefore purely serves as an indicator for the reliability of the SHAP assessment and is not a measure of out-of-sample predictive performance. However, [Table 1](#) clearly shows that XGBoost provides a much better in-sample R^2 than the linear panel regression as it enables us to capture potential non-linear dynamics.

Table 1

Empirical Findings. The table presents the coefficients of the fixed effect panel regression (FE) and the average Shapley Values of the XG-Boost exercise (XGB + Shap) for each explanatory variable.

Target variable Sample	$h = 3$ Full sample		$h = 4$ Full sample		$h = 3$ Advanced Economies		$h = 3$ Emerging Economies	
	FE	XGB + Shap	FE	XGB + Shap	FE	XGB + Shap	FE	XGB + Shap
MP_Shock_Diff	1.111	2.700	0.460	3.454	-5.586*	5.569	1.339*	4.237
GDP_Forecast_Diff	0.003	1.448	-1.086	1.560	-0.079	1.399	0.270	2.686
GDP_Forecast_Error	-0.410	3.668	-0.158	2.597	-0.902**	2.093	-0.132	4.059
CPI_Forecast_Diff	0.092	2.002	-1.040*	2.751	-0.643	1.486	0.011	2.586
CPI_Forecast_Error	1.488**	3.668	1.865**	4.158	-0.541	1.899	1.472**	8.223
WUI_Shock_Diff	-1.039	2.791	2.341	3.025	-2.814	2.108	-1.338	3.741
Debt_to_GDP	0.163*	2.102	0.028	2.714	0.064	0.667	0.289	3.741
CBI	-29.259	0.401	-33.115	1.615	6.692	0.405	-37.517	0.688
N	406	406	406	406	182	182	224	224
R ²	0.162	0.989	0.206	0.999	0.184	0.999	0.208	0.999
RMSE	11.359	1.930	10.123	0.126	6.871	0.029	13.096	0.030

Target describes the target variable.

* p -value below 5%.

** p -value below 1%

role for exchange rate expectations in EMDE compared to economies where inflation rates (and therefore also their forecast errors) have been modest.

Monetary policy shocks (MP_Shock_Diff) exert a significantly negative effect for AE and a significantly positive effect for EMDE (fixed-effect: -5.586 for AE and 1.339 for EMDE). The latter is consistent with uncovered interest parity (UIP) and indicates that a stronger domestic monetary policy shock compared to the US is associated with depreciation expectations, while the former indicates a violation of UIP in line with previous literature (Engel, 2014). However, for the full sample, the association of MP_Shock_Diff with revisions in exchange rate expectations is not significant and potentially vanishes due to offsetting effects. In our view, a key strength of the Shapley value approach is that it is able to reveal meaningful variable contributions even when traditional (pooled) regression results are not significant. Specifically, in our full sample, MP_Shock_Diff shows no statistical significance due to opposing dynamics between AE and EMDE, which may cancel each other out. This finding in combination with the concern that splitting the sample a priori imposes a discrete form of heterogeneity, which may not always be clear-cut, underlines the benefit of the Shapley value approach. The SHAP values still highlight monetary policy shocks as main determinant of exchange rate expectations (second largest SHAP value for $h = 4$ and fourth place for $h = 3$).

Similarly, economic uncertainty shocks (WUI_Shock_Diff) seem to matter across all samples according to the relatively high SHAP values, despite their insignificance in fixed-effects regressions. Interestingly, economic uncertainty shocks receive the highest SHAP value for EMDE, suggesting that risk-related factors are particularly relevant in these markets. This implies that uncertainty may affect expectations in a nonlinear, state-dependent fashion, becoming relevant in high-risk environments typical for emerging markets. However, the interpretation of uncertainty effects must take into account the horizon of expectations. While higher uncertainty is typically associated with safe-haven flows and a short-run depreciation of riskier currencies, our survey data capture long-term expectations. At such horizons, professional forecasters may anticipate a reversal of short-run deviations driven by uncertainty, reflecting mean reversion toward fundamentals such as purchasing power parity or the temporary nature of policy-related uncertainty shocks. This would explain the negative sign in fixed effects regressions. As a result, increases in relative uncertainty may be associated with expected appreciation over longer horizons, even if the short-run effect points in the opposite direction.

GDP growth forecast errors display statistically significant negative coefficients in the fixed-effects regression for AE (-0.902), while exhibiting relatively similar importance according to SHAP values across all samples. Professional forecasters tend to anticipate an appreciation of the domestic currency when shocks to GDP growth are stronger in the domestic economy than in the US. The negative association between relative GDP growth forecast errors and exchange rate expectations can be interpreted through multiple channels. While the monetary model suggests that stronger domestic growth increases money demand and leads to an appreciation, alternative explanations are also consistent with our findings. In particular, higher relative growth may signal improved investment opportunities and higher expected returns, attracting capital inflows and strengthening the currency. GDP growth forecast errors seem to matter solely for exchange rate expectations in AE, when referring to the statistical significance of the fixed effects regressions. However, when comparing the Shapley values, it becomes evident that the contribution of GDP growth forecast errors for exchange rate expectations might be even larger for EMDE. This also demonstrates the added value of XGBoost with Shapley values as a tool for uncovering such nuanced, non-linear relationships, which is particularly beneficiary in our context due to its complementarity to the regression framework.

In contrast, the CBI index shows a statistically insignificant association in the fixed-effects regression and contributes almost nothing in the XGBoost model. This may suggest an indirect transmission mechanism: central bank independence might operate through other variables (e.g., MP_Shock_Diff) rather than exerting a direct influence.

Fig. 1 illustrates the individual SHAP values for each (sub-)sample and allows us to gain further insights into the relevance of each determinant. In addition to its prominent ranking, we find that the positive impact from CPI_Forecast_Error at $h = 3$ in EMDE

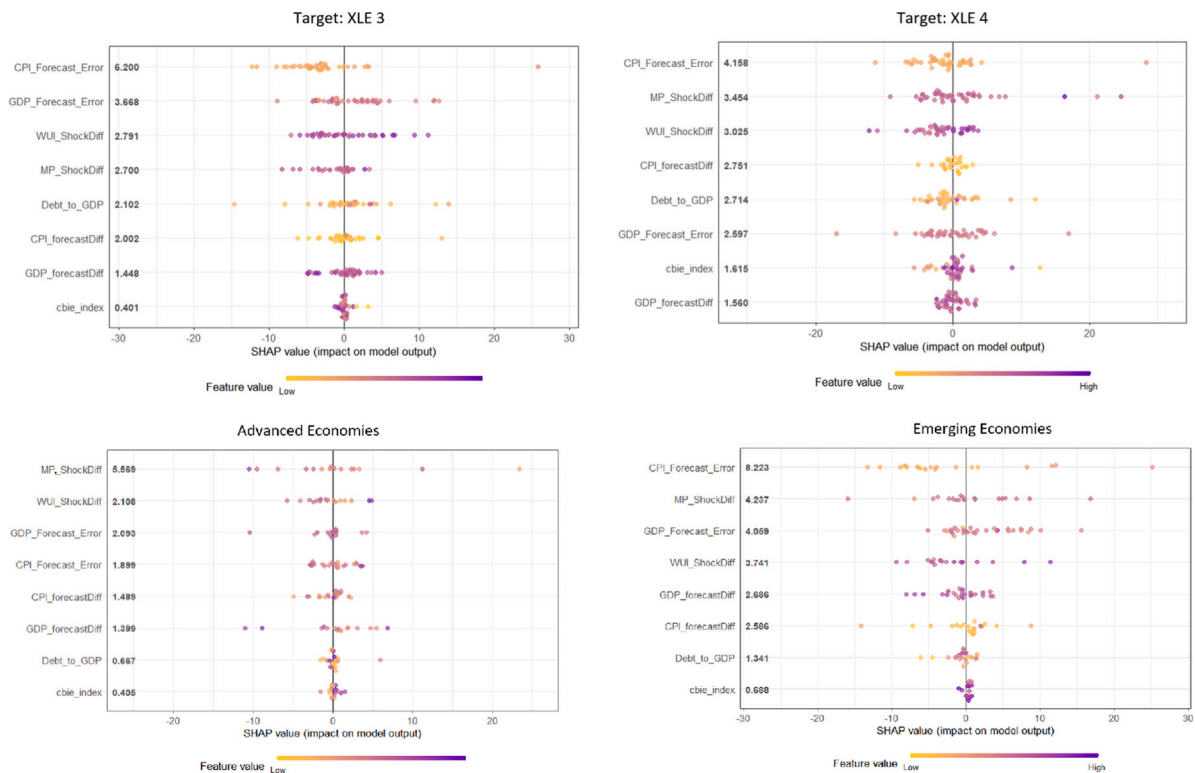


Fig. 1. Explainable AI. This figure depicts both local and global importance of the determinants, quantified using SHAP values for the four sub-samples defined in Table 1. The upper left panel shows long-term exchange rate expectations with a 3-year horizon (XLE3) for the full sample, while the upper right panel presents the corresponding 4-year horizon (XLE4). The lower panels display 3-year exchange rate expectations for Advanced Economies (left) and Emerging Economies (right), respectively. The x-axis represents the local SHAP value of each observation, indicating its contribution to the predicted outcome, whereas the y-axis reports the mean absolute SHAP value, capturing the global importance of each determinant. Marker colors reflect the underlying feature values of the individual observations.

is confirmed by local SHAP explanation. Low values of inflation forecast errors (indicated in yellow) are described with low SHAP values. The local explanations for *MP_ShockDiff*, which is ranked as the second most important determinant, show that higher feature values (colored purple) correspond to larger SHAP values.

In general, the consistency between the fixed effect model and XGBoost is high and strengthens the credibility of the findings. In contrast, discrepancies for some variables underscore the potential importance of nonlinear modeling approaches. The XGBoost model achieves an excellent in-sample fit, indicated by very high R^2 s (0.999) and very low RMSEs (0.029–0.126). Subgroup analysis (AE vs. EMDE) reveals notable differences: For AE, monetary policy seems to be the dominant driver. For EMDE, inflation and GDP growth forecast errors are key determinants. Overall, changes in exchange rate expectations are driven not only by macroeconomic fundamentals but primarily by inflation forecast errors, uncertainty shocks, and monetary policy shocks. Combining traditional econometric methods with modern machine learning techniques provides deeper insights into expectation formation and reveals substantial heterogeneity across different groups of economies. In particular in case where regression coefficients go into opposite directions across sub-groups (AE vs. EMDE) in traditional regression models, the XGBoost model seems to be better suited to capture non-linear relationships and interaction effects that are not reflected in the linear fixed effects specification.

While differences between advanced and emerging economies are well documented, splitting the sample a priori imposes a discrete form of heterogeneity and the distinction between advanced and emerging market economies may not always be clear-cut for each country. In contrast, our pooled approach combined with explainable machine learning allows for continuous, state-dependent, and interaction-driven heterogeneity. This is particularly useful as the relevance of key drivers such as monetary policy and uncertainty may depend not only on country group but also on macroeconomic conditions such as inflation regimes or risk environments. By exploiting the full cross-country variation, the model is able to allow for nonlinear patterns that would not be addressed in separate estimations. Consequently, we see the two approaches as complementarity to each other. While SHAP values provide insights into the relative importance and conditional contribution to prediction accuracy of different variables, they do not identify structural parameters. The interpretation of the results therefore relies on economic reasoning and should be understood as indicative of underlying mechanisms rather than definitive causal relationships.

Table A.1

Descriptive Statistics. The table presents descriptive statistics for the target and the explanatory variables from 2008–2021. N is the amount of observations, Min and Max are the minimum and maximum value of each variable. Median, Q1 and Q3 are the corresponding quartiles of the empirical return distribution. IQR is the interquartile range, MAD is the median absolute deviation, SD the standard deviation of the mean, SE the standard error of the mean and CI is the 95% confidence interval of the mean.

2008–2021	N	Min	Max	Median	Q1	Q3	IQR	MAD	Mean	SD	SE	CI
Full sample												
Target variables												
FX forecast revision ($h = 3$)	406	−33.74	97.41	2.88	−3.75	12.88	16.63	11.75	5.96	17.51	0.87	1.71
FX forecast revision ($h = 4$)	406	−50.72	97.46	4.88	−1.53	11.85	13.39	9.83	6.82	16.85	0.84	1.64
Explanatory variables												
MP_Shock_Diff	406	−12.84	10.71	−0.03	−0.49	0.66	1.15	0.83	0.06	1.60	0.08	0.16
GDP_Forecast_Diff	406	−12.40	9.30	0.20	−0.70	1.30	2.00	1.48	0.55	2.08	0.10	0.20
GDP_Forecast_Error	406	−15.72	17.69	−0.07	−1.40	0.89	2.29	1.65	−0.37	2.92	0.14	0.28
CPI_Forecast_Diff	406	−2.20	45.60	0.80	−0.20	2.30	2.50	1.78	1.90	4.57	0.23	0.45
CPI_Forecast_Error	406	−8.11	30.75	−0.11	−0.94	0.89	1.83	1.32	0.36	3.21	0.16	0.31
WUI_Shock_Diff	406	−1.82	0.89	0.02	−0.22	0.25	0.47	0.35	0.01	0.35	0.02	0.04
Debt_to_GDP	406	4.92	226.08	40.69	32.12	69.01	36.89	18.41	53.24	35.84	1.78	3.50
CBI	406	0.26	0.95	0.64	0.50	0.76	0.26	0.19	0.62	0.16	0.01	0.02
Advanced Economies												
Target variables												
FX forecast revision ($h = 3$)	182	−31.53	79.16	1.73	−3.91	6.11	10.02	8.09	2.64	12.37	0.92	1.81
FX forecast revision ($h = 4$)	182	−30.32	79.51	4.46	−0.97	8.97	9.94	7.72	4.48	11.19	0.83	1.64
Explanatory variables												
MP_Shock_Diff	182	−2.35	3.16	−0.08	−0.44	0.41	0.85	0.58	0.05	0.78	0.06	0.11
GDP_Forecast_Diff	182	−2.70	4.00	−0.30	−0.90	0.30	1.20	0.89	−0.20	1.11	0.08	0.16
GDP_Forecast_Error	182	−11.40	10.72	0.01	−1.15	0.70	1.85	1.23	−0.31	2.37	0.17	0.35
CPI_Forecast_Diff	182	−2.20	3.80	0.00	−0.57	0.50	1.07	0.81	0.05	0.99	0.07	0.14
CPI_Forecast_Error	182	−4.03	4.75	−0.19	−0.87	0.46	1.33	1.00	−0.06	1.33	0.10	0.19
WUI_Shock_Diff	182	−0.80	0.89	0.02	−0.21	0.21	0.42	0.32	0.00	0.32	0.02	0.05
Debt_to_GDP	182	11.29	226.08	42.07	35.66	86.55	50.89	17.54	64.67	46.05	3.41	6.74
CBI	182	0.37	0.95	0.60	0.45	0.68	0.23	0.22	0.59	0.16	0.01	0.02
Emerging Economies												
Target variables												
FX forecast revision ($h = 3$)	224	−33.74	97.41	4.47	−3.52	17.23	20.75	14.07	8.65	20.41	1.36	2.69
FX forecast revision ($h = 4$)	224	−50.72	97.46	6.10	−2.98	15.34	18.32	13.73	8.71	20.15	1.35	2.65
Explanatory variables												
MP_Shock_Diff	224	−12.84	10.71	0.01	−0.55	0.80	1.35	0.98	0.08	2.04	0.14	0.27
GDP_Forecast_Diff	224	−12.40	9.30	0.90	−0.30	2.80	3.10	2.08	1.16	2.45	0.16	0.32
GDP_Forecast_Error	224	−15.72	17.69	−0.12	−1.54	1.00	2.54	1.96	−0.42	3.31	0.22	0.44
CPI_Forecast_Diff	224	−2.10	45.60	2.00	0.90	4.00	3.10	2.08	3.40	5.67	0.38	0.75
CPI_Forecast_Error	224	−8.11	30.75	−0.02	−1.07	1.46	2.53	1.83	0.70	4.13	0.28	0.54
WUI_Shock_Diff	224	−1.82	0.87	0.02	−0.23	0.28	0.51	0.38	0.01	0.38	0.03	0.05
Debt_to_GDP	224	4.92	103.00	39.57	28.80	57.14	28.34	18.28	43.96	20.45	1.37	2.69
CBI	224	0.26	0.85	0.66	0.56	0.77	0.21	0.15	0.65	0.15	0.01	0.02

5. Concluding remarks

This paper revisits the determinants of long-term exchange rate expectations by combining traditional panel regressions with explainable machine learning techniques. Although linear models confirm the central role of inflation forecast errors consistent with PPP, they provide limited evidence for the relevance of monetary policy and uncertainty shocks. By contrast, the XGBoost framework complemented with SHAP values reveals that both monetary policy and uncertainty shocks play an important role once nonlinearities and interactions are taken into account. In particular, uncertainty emerges as a key determinant in emerging economies, pointing to the importance of state-dependent and risk-related mechanisms in expectation formation. From a methodological perspective, our findings demonstrate that explainable machine learning can serve as a valuable complement to standard econometric approaches. It allows to uncover hidden relationships while retaining a degree of interpretability.

Our findings offer several implications. For policymakers, they suggest that the transmission of monetary policy to exchange rate expectations is more complex and state-dependent than implied by linear models. For emerging markets, in particular, managing inflation and reducing uncertainty appear crucial to stabilizing exchange rate expectations. For researchers, the results underline the importance of integrating flexible, data-driven methods with economic theory to better understand expectation formation in international financial markets. Future research could extend this analysis by evaluating out-of-sample predictive performance.

CRediT authorship contribution statement

Theo Berger: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Robert L. Czudaj:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Reyhaneh Rezaei Khavas:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See [Table A.1](#).

Data availability

Data will be made available on request.

References

- Beckmann, J., Czudaj, R.L., 2025. Fundamental determinants of exchange rate expectations. *Int. J. Forecast.* 41, 1003–1021.
- Berger, T., 2023. Explainable artificial intelligence and economic panel data: A study on volatility spillover along the supply chains. *Financ. Res. Lett.* 54, 103757.
- Berger, T., 2025. On the information content of explainable artificial intelligence for quantitative approaches in finance. *OR Spectrum* 1436–6304, 177–203.
- Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. KDD'16, ACM, pp. 785–794.
- Czudaj, R.L., Rezaei Khavas, R., 2026. Exchange rate expectations over longer horizons and the role of monetary policy and uncertainty. *Emerg. Mark. Rev.* forthcoming, <https://drive.google.com/file/d/1cCljRu5-Welz3rDktrbT3GT8LtOJOC3P/view?usp=sharing>.
- Dornbusch, R., 1976. Expectations and exchange rate dynamics. *J. Political Econ.* 84, 1161–1176.
- Eichenbaum, M., Evans, C., 1995. Some empirical evidence on the effects of shocks to monetary policy on exchange rates. *Q. J. Econ.* 110, 975–1009.
- Engel, C., 1996. The forward discount anomaly and the risk premium: A survey of recent evidence. *J. Empir. Financ.* 3, 123–192.
- Engel, C., 2014. Exchange rates and interest parity. In: *Handbook of International Economics*, vol. 4, Elsevier, pp. 453–522, chap. Chapter 8.
- Frankel, J., Chinn, M., 1993. Exchange rate expectations and the risk premium: Tests for a cross section of 17 currencies. *Rev. Int. Econ.* 1, 136–144.
- Furceri, D., Loungani, P., Zdzienicka, A., 2018. The effects of monetary policy shocks on inequality. *J. Int. Money Financ.* 85, 168–186.
- Habib, M.M., Stracca, L., 2012. Getting beyond carry trade: What makes a safe haven currency? *J. Int. Econ.* 87, 50–64.
- Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions. *Adv. Neural Inf. Process. Syst.* 30, 4765–4774.
- Meese, R.A., Rogoff, K., 1983. Empirical exchange rate models of the seventies: Do they fit out of sample? *J. Int. Econ.* 14, 3–24.
- Müller, G.J., Wolf, M., Hettig, T., 2024. Delayed overshooting: The case for information rigidities. *Am. Econ. J.: Macroecon.* 16, 310–342.
- Ranaldo, A., Söderlind, P., 2010. Safe haven currencies. *Rev. Financ.* 14, 385–407.
- Romelli, D., 2022. The political economy of reforms in Central Bank design: Evidence from a new dataset. *Econ. Policy* 37, 641–688.
- Rossi, B., 2013. Exchange rate predictability. *J. Econ. Lit.* 51, 1063–1119.